

EMILIANA PRATES

CONTACT

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- Portugal

SKILLS

Core Competencies

- Full-Stack Development
- Backend Development
- Android Development
- Game Development
- Cybersecurity

Tools & Frameworks

- Node.js
- Express.js
- Spring
- Docker
- MySQL
- SQLite
- Git
- GitHub

Programming Languages

- Java
- TypeScript
- JavaScript
- Python
- Kotlin
- C#
- C++
- PHP
- HTML5
- CSS3

Additional Technologies

- Arduino
- Unity
- Blender
- Adobe Illustrat

LANGUAGES

- Portuguese (Native)
- English (Fluent Proficiency)

HOBBIES & INTERESTS

- Digital Art
- 3d Modeling
- Low-Level Programming
- Physical Computing



PROFILE

Informatics and Multimedia Engineering graduate and aspiring software engineer. I enjoy working across the stack and moving between different kinds of problems, from designing backend systems and web applications to building mobile experiences and experimenting with game development.



EDUCATION

08/2023 - 07/2026 Lisbon, Portugal

BSc Informatics and Multimedia Engineering

Lisbon School of Engineering | Polytechnic University of Lisbon

Relevant Coursework: Algorithms & Programming · Object-Oriented Programming · Operating Systems · Computer Networks · Distributed Systems · Database Systems · Web Development · Mobile Applications · Artificial Intelligence · Machine Learning · Image Processing

Electives: Cybersecurity · Graphic Design · Interaction in Virtual Environments

Website: www.isel.pt/en

Final Grade: 14 / 20



PROJECTS

Course Certificate Generation System 02/2026 - 07/2026

University Capstone Project

TypeScript · Node.js · Express.js · MySQL · Docker · Casbin

Developed a full-stack application to automate certificate generation, digital signing and email delivery.

Implemented role-based access control using Casbin and authentication with a MySQL database.

Integrated academic databases to retrieve and validate student and course information.

Implemented customizable certificate templates and cryptographic document signing.

[View on GitHub](#)

Ant Colony and Pheromone Simulator 11/2025 - 01/2026

Java · Processing · Agent-Based Simulation

Developed an agent-based simulation modeling ant behavior.

Implemented individual agent behavior and indirect communication through pheromone trails, producing emergent colony behavior.

Built graphical visualization in Processing and integrated procedurally generated audio based on ant movement.

[View on GitHub](#)

Arduino Gyroscope 10/2023 - 01/2024

C++ · Arduino · Embedded Systems

Developed an embedded application to capture and display X/Y/Z accelerometer data in real time and convert it to tilt.

Implemented I2C communication with an external LCD and serial data transmission over USB.

Controlled the hardware directly without external sensor libraries.

[View on GitHub](#)